

Notes: Modigliani and Miller (AER, 1958)

The Cost of Capital, Corporation Finance and the Theory of Investment

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1 Big picture

- **Question.** In a world with uncertain corporate earnings, what is the firm's cost of capital, and can managers raise firm value by changing the mix of debt and equity?
- **Setting.** Firms own risky assets that generate uncertain perpetual income streams. Securities are traded in perfect capital markets, and firms can be grouped into risk classes with comparable underlying asset risk.
- **Core challenge.** Under uncertainty, "maximize profits" is no longer operational because profit is a random variable. The paper therefore replaces the profit-maximization approach with a market-value approach.
- **Main conceptual contribution.** The paper uses an arbitrage argument to show that financial structure is irrelevant for firm value in perfect capital markets. This is the classic M-M irrelevance result.
- **Proposition I.** For any firm j in risk class k , firm value satisfies

$$V_j = S_j + D_j = \frac{X_j}{p_k}, \quad (1)$$

where X_j is expected operating income before interest. Hence the market value of the firm is independent of leverage.

- **Equivalent cost-of-capital statement.** The average cost of capital is

$$\frac{X_j}{V_j} = p_k, \quad (2)$$

so it depends only on the asset-risk class, not on capital structure.

- **Proposition II.** Equity becomes riskier when debt rises, so the expected yield on common stock increases linearly with leverage:

$$i_j = p_k + (p_k - r) \frac{D_j}{S_j}. \quad (3)$$

Debt does not create value; it just reallocates risk from the whole firm to equity.

- **Proposition III.** Investment policy is also financing-irrelevant in the baseline model. A project with expected return p^* should be undertaken if and only if

$$p^* \geq p_k, \quad (4)$$

regardless of whether the project is financed by debt, retained earnings, or a stock issue.

- **Tax extension.** Once corporate interest deductibility is introduced, debt can create value through a tax shield, but the paper argues the quantitative gain is much smaller than standard finance doctrine suggested.
- **Empirical contribution.** The paper presents preliminary cross-sectional evidence from electric utilities and oil companies. The evidence is broadly consistent with a flat average cost of capital and a positive relation between equity yields and leverage.
- **Economic takeaway.** In perfect capital markets, financing policy does not change the size of the pie. It only changes how the pie is sliced among claimholders.

2 Data and measurement (key objects)

2.1 Why this is mainly a theory paper

- The paper is primarily theoretical. There is no modern causal design, no instrument, and no structural estimation of a dynamic model.
- The key empirical objects are therefore the model primitives and valuation variables, plus a small set of summary ratios used in the paper's preliminary evidence section.
- The most important objects are:
 - expected operating income before interest, X_j ,
 - market value of debt, D_j ,
 - market value of common equity, S_j ,
 - total firm value, $V_j = S_j + D_j$,
 - the risk-class capitalization rate, p_k ,
 - the safe interest rate, r ,
 - the equity yield, i_j ,
 - and the project return p^* used in the investment rule.

Interpretation. The paper is not measuring treatment effects; it is defining the valuation objects that determine whether leverage matters.

2.2 Uncertain return streams

Each firm's assets generate a stream of uncertain profits over time. Let $X_i(t)$ denote profit in period t . The paper defines the firm's long-run average return as

$$X_i = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T X_i(t), \quad (5)$$

with expected value

$$\bar{X}_i = \mathbb{E}[X_i]. \quad (6)$$

If the firm has N_i shares outstanding, the expected return per share is

$$\bar{x}_i = \frac{\bar{X}_i}{N_i}. \quad (7)$$

Interpretation. The paper does not assume certainty-equivalent cash flows. Instead, it works directly with risky income streams and asks how markets capitalize them.

2.3 Risk classes

The paper assumes that firms can be partitioned into homogeneous *risk classes*. Within a class, returns are proportional across firms, so securities differ only by a scale factor once underlying asset risk is normalized.

For any firm j in class k , the price per dollar of expected return is the same. Thus,

$$p_j = \frac{x_j}{p_k}, \quad (8)$$

or equivalently,

$$\frac{x_j}{p_j} = p_k. \quad (9)$$

Interpretation. The class concept is the paper’s substitute for “same risk.” It allows the model to compare firms with different leverage but similar underlying asset risk.

2.4 Core balance-sheet and valuation objects

The paper’s central balance-sheet decomposition is

$$V_j = S_j + D_j, \quad (10)$$

where:

- S_j is the market value of common stock,
- D_j is the market value of debt,
- V_j is total market value.

The main income concept is expected operating return before interest, X_j . The two most important ratios in the theory are therefore

$$\frac{X_j}{V_j} \quad \text{and} \quad \frac{D_j}{S_j}. \quad (11)$$

Interpretation. The first ratio is the average cost of capital in the baseline model. The second is the measure of leverage that drives the cost of equity.

2.5 The bond-market object

In the baseline leverage model, all bonds are treated as perfect substitutes and earn the same sure rate of return r .

Interpretation. This assumption is crucial. It lets an investor borrow or lend on personal account at the same rate as a corporation, which is exactly what makes “homemade leverage” possible.

2.6 What is “measured” in the preliminary evidence section

The paper’s empirical section is very modest, but it does define observable proxies for the key theoretical objects.

- **Electric utilities sample.** 43 large electric utilities, using average data for 1947–1948.
- **Oil sample.** 42 oil companies, using 1953 data.

The paper constructs two main empirical tests.

- For Proposition I, it studies an average-cost proxy

$$x \approx \frac{\text{net return}}{V}, \quad (12)$$

against leverage measured by

$$d = \frac{D}{V}. \quad (13)$$

- For Proposition II, it studies an equity-yield proxy

$$z \approx \frac{\text{net stockholder income}}{S}, \quad (14)$$

against leverage measured by

$$h = \frac{D}{S}. \quad (15)$$

Interpretation. The empirical section does not identify causality. It simply checks whether cross-sectional patterns line up with the paper’s comparative-statics predictions.

3 Models and empirical specifications (all equations + interpretation)

3.1 Baseline valuation without leverage

Start with an all-equity firm in class k . Since all securities in that class sell at the same price per dollar of expected return,

$$V_j = S_j = \frac{X_j}{p_k}. \quad (16)$$

Interpretation. In the pure-equity benchmark, p_k is both the capitalization rate for the risky asset stream and the firm’s average cost of capital.

3.2 Proposition I: capital structure irrelevance

Once firms can issue debt, the paper’s main claim is:

$$V_j = S_j + D_j = \frac{X_j}{p_k}, \quad \text{for any firm } j \text{ in class } k. \quad (17)$$

Equivalently,

$$\frac{X_j}{V_j} = p_k. \quad (18)$$

Interpretation. Leverage changes the division of cash flows between debt and equity, but it does not change the total present value of the underlying risky operating stream.

3.3 Arbitrage proof intuition for Proposition I

Take two firms in the same class with identical operating income X , but different leverage.

- If the levered firm is *overvalued*, an investor can sell the levered firm's equity and replicate the same payoff by buying unlevered equity and borrowing personally.
- If the levered firm is *undervalued*, an investor can “undo” the leverage by buying the levered firm's stock and bonds in the right proportions.

Because investors can lever or unlever on personal account, any gap in total value between levered and unlevered firms in the same class generates a pure arbitrage opportunity.

Interpretation. The result is not driven by a specific utility function. It is driven by a no-free-lunch replication argument.

3.4 Proposition II: leverage raises the cost of equity

Define the expected return on equity as

$$i_j = \frac{X_j - rD_j}{S_j}. \quad (19)$$

Using Proposition I, the paper shows

$$i_j = p_k + (p_k - r) \frac{D_j}{S_j}. \quad (20)$$

Interpretation. Equityholders require a higher expected return when leverage rises because leverage concentrates the firm's residual risk onto equity. This is why debt is not “cheap” in an overall sense even if the bond rate r is below the equity yield.

3.5 The paper's direct challenge to traditional doctrine

Traditional corporate-finance doctrine effectively assumed that, over a moderate leverage range, the equity capitalization rate stays roughly constant while debt remains cheap. In the paper's notation, that view implies a falling average cost of capital as leverage rises.

M–M rejects exactly this logic. Once the equity yield adjusts endogenously with leverage, the average cost of capital remains flat in the no-tax benchmark.

Interpretation. The whole paper is a critique of the idea that managers can create value merely by substituting low-yield debt for high-yield equity.

3.6 Tax-adjusted Proposition I

When corporate interest is deductible, the relevant after-tax income measure becomes

$$X_j^T = (X_j - rD_j)(1 - T) + rD_j, \quad (21)$$

where T is the corporate income tax rate.

The tax-adjusted irrelevance statement is then

$$\frac{X_j^\tau}{V_j} = p_k^\tau. \quad (22)$$

Interpretation. Taxes break the original version of strict irrelevance because interest deductibility makes debt more valuable. But the paper stresses that the gain is a tax-shield effect, not a generic cost-of-capital miracle.

3.7 Tax-adjusted Proposition II

The cost-of-equity formula becomes

$$i_j = p_k^\tau + (p_k^\tau - r) \frac{D_j}{S_j}. \quad (23)$$

Interpretation. Even with corporate taxes, leverage still raises the required return on equity. Debt does not erase risk; it just changes how that risk is priced.

3.8 Rising borrowing rates and other extensions

The paper also considers the case where the borrowing rate rises with leverage. In that case:

- Proposition I still survives in spirit if the same borrowing schedule applies symmetrically to arbitrageurs and firms.
- Proposition II need not stay exactly linear; the equity-yield curve can bend as debt becomes more expensive at high leverage.

Interpretation. The clean linear formula is a special case. The deeper result is the arbitrage logic behind total-value irrelevance.

3.9 Proposition III: the investment rule

Let an investment of size I yield expected return p^*I .

If financed by debt, pre-investment and post-investment values satisfy

$$V_0 = \frac{X_0}{p_k}, \quad V_1 = \frac{X_0 + p^*I}{p_k}. \quad (24)$$

Since new debt of size I is added,

$$S_1 = V_1 - (D_0 + I) = S_0 + \left(\frac{p^*}{p_k} - 1 \right) I. \quad (25)$$

So

$$S_1 \geq S_0 \iff p^* \geq p_k. \quad (26)$$

The paper shows the same cut-off for retained earnings and new stock issues as well.

Interpretation. Financing choice drops out of the investment rule. The project is good if its return beats the class capitalization rate.

3.10 Tax-adjusted investment cutoffs

With taxes, the cut-off can differ by financing source. The paper derives:

$$p_k^S = \frac{p_k^\tau}{1 - T} \quad (27)$$

for new common stock,

$$p_k^D = \frac{p_k^\tau}{1 - T} \left(1 - \frac{rT}{p_k^\tau} \right) \quad (28)$$

for pure debt financing, and

$$p_k^R = \frac{1 - T_d}{1 - T_g} \cdot \frac{p_k^\tau}{1 - T} \quad (29)$$

for retained earnings, where T_d is the personal tax on dividends and T_g is the personal tax on capital gains.

Interpretation. Taxes generate financing wedges, but the paper argues these wedges are much smaller than the finance literature had typically assumed.

4 Identification and instruments (what makes the estimates credible)

4.1 No causal identification design

- The paper does not identify a reduced-form treatment effect.
- There is no instrument, no panel fixed-effects design, and no quasi-experimental variation.
- The paper’s credibility comes instead from the transparency of its arbitrage logic and the sharpness of its assumptions.

Interpretation. The relevant question is not “what identifies a coefficient?” but “which assumptions make homemade leverage possible?”

4.2 The core identifying logic is replication

The model’s main conclusions come from the fact that an investor can reproduce a corporate leverage choice in personal portfolios.

- If the corporation issues more debt, investors can emulate the same leverage by borrowing personally.
- If the corporation is levered and an investor dislikes that structure, the investor can unbundle the firm’s leverage by holding the stock and the bonds in the right proportions.

Interpretation. Proposition I is really a spanning result. If corporate leverage adds no payoff that households cannot synthesize on their own, it cannot command a valuation premium.

4.3 The assumptions doing the heavy lifting

1. **Perfect capital markets.** No transaction costs, no meaningful borrowing frictions, and competitive pricing.
2. **Symmetric borrowing access.** Households and firms can borrow at the same rate in the baseline argument.
3. **Homogeneous risk classes.** Firms in the same class differ only in scale and leverage, not in underlying asset risk.
4. **No bankruptcy or distress costs.** Debt does not destroy value through real resource costs.
5. **No asymmetric information or agency frictions.** Financing does not signal private information and does not distort managerial behavior in the baseline model.
6. **Static partial equilibrium.** The paper takes capitalization rates as given rather than jointly determining them in general equilibrium.

Interpretation. The irrelevance theorem is exact only under these strong conditions. The point of the paper is to make those conditions explicit and then show what follows from them.

4.4 Why taxes matter

The cleanest break from irrelevance is the corporate tax shield. Because interest is deductible at the corporate level, leverage can raise after-tax firm value.

Interpretation. This is not a contradiction of the theorem. It shows exactly what kind of market imperfection is required to make capital structure matter.

4.5 What the preliminary evidence can and cannot establish

- The empirical section is a suggestive consistency check, not a decisive test.
- The paper uses noisy proxies for expected returns, so measurement error is substantial.
- Industrial classification is crude, especially in the oil sample, so the risk-class assumption only holds approximately.
- The evidence is cross-sectional and static, so it cannot separately identify taxes, bankruptcy costs, or institutional frictions.

Interpretation. The empirical evidence is best viewed as “nothing obvious in the data contradicts the theorem,” not as a modern clean validation exercise.

4.6 Where the theorem can fail in practice

The paper itself flags several reasons why real-world results may depart from the benchmark.

- institutional portfolio restrictions,

- unequal borrowing rates between households and firms,
- taxes,
- other market imperfections that interfere with arbitrage,
- and dynamic issues the static partial-equilibrium model leaves aside.

Interpretation. M–M is a benchmark, not a claim that capital structure never matters in practice.

4.7 Relation to later corporate-finance logic

Within the paper itself, the contrast is mainly against older “cheap debt” doctrine. The paper’s lasting contribution is methodological: start from the no-frictions benchmark, then add specific frictions one by one.

Interpretation. That move is what made the modern trade-off and market-imperfection approach to capital structure possible.

5 Tables and figures

This paper mixes conceptual diagrams with a small amount of preliminary cross-sectional evidence. The cleanest way to read the exhibits is to separate *theoretical prediction maps* from *empirical consistency checks*.

5.1 Figure 1: average cost of capital versus leverage

Read. The paper’s conceptual Figure 1 contrasts two views.

- The traditional doctrine predicts that the firm’s average cost of capital falls as leverage rises, at least over a moderate range.
- M–M predicts a horizontal line: the average cost of capital is flat with respect to leverage in the no-tax benchmark.

Economic message. This is the paper’s central disagreement with conventional corporate-finance wisdom.

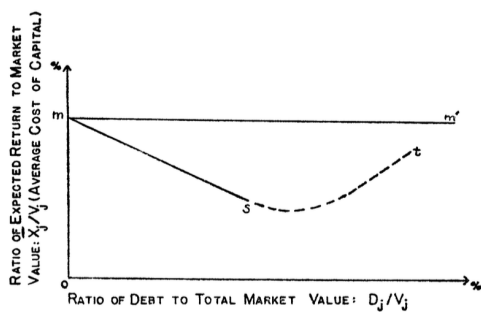


FIGURE 1

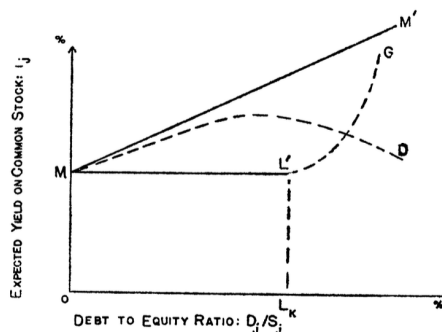


FIGURE 2

5.2 Figure 2: equity yield versus leverage

Read. The conceptual Figure 2 contrasts a nearly flat traditional equity-yield schedule with the M–M prediction that common-stock yields rise with leverage.

Economic message. The cost of equity is the adjustment margin that keeps the firm’s overall cost of capital from falling when cheap debt is added.

5.3 Exhibit 1: proposition map

Table 1: What the main propositions establish

Proposition	Formal statement	Economic meaning
I	$V_j = X_j/p_k$ and $X_j/V_j = p_k$ for any firm in class k .	Total firm value and the average cost of capital are independent of leverage in perfect capital markets.
II	$i_j = p_k + (p_k - r)D_j/S_j$.	Leverage raises the required return on equity because it makes equity riskier.
III	Invest if and only if $p^* \geq p_k$, regardless of financing source.	Financing policy does not alter the investment cutoff in the no-tax benchmark.

Read. The three propositions move in a sequence: valuation irrelevance, equity-yield adjustment, and investment irrelevance.

Economic message. The paper is not only about capital structure. It is equally a theory of the correct hurdle rate for investment.

5.4 Exhibit 2: traditional doctrine versus M–M

Table 2: Competing predictions in the paper

Object	Traditional view criticized by the paper	M–M benchmark prediction
Average cost of capital	Falls with leverage over a moderate range because debt is “cheap.”	Stays constant with leverage in the no-tax benchmark.
Cost of equity	Roughly flat over moderate leverage; rises only at very high debt ratios.	Rises immediately and approximately linearly with leverage.
Firm value	Can be increased by replacing equity with debt.	Unchanged by capital structure in perfect markets.
Investment cutoff	Can depend on financing source because debt is supposedly cheaper.	Depends only on asset-risk class, not on financing source.

Read. The paper is a direct theoretical rebuttal to the “substitute cheap debt for expensive equity” logic.

Economic message. Once equilibrium pricing of equity is taken seriously, cheap debt does not create free value.

5.5 Exhibit 3: preliminary evidence on Proposition I

Table 3: Cross-sectional tests of average cost of capital versus leverage

Sample	Dependent variable	vari-	Reported relation	Read
43 electric utilities, 1947–48	$x \approx$ net return/ V		$x = 5.3 + 0.006d$ with $r = 0.12$	The slope is near zero, not negative as the traditional view predicts.
42 oil companies, 1953	$x \approx$ net return/ V		$x = 8.5 + 0.006d$ with $r = 0.04$	Again, no evidence that average cost of capital declines with leverage.

Read. The paper’s first empirical check is whether the proxy for X^τ/V falls with D/V . It does not.

Economic message. The early data do not support the textbook claim that leverage mechanically lowers the firm’s overall cost of capital.

5.6 Exhibit 4: preliminary evidence on Proposition II

Table 4: Cross-sectional tests of equity yield versus leverage

Sample	Dependent variable	vari-	Reported relation	Read
43 electric utilities, 1947–48	z net stockholder income/ S	$\approx$	$z = 6.6 + 0.017h$ with $r = 0.53$	Common-stock yields rise with leverage, as Proposition II predicts.
42 oil companies, 1953	z net stockholder income/ S	$\approx$	$z = 8.9 + 0.051h$ with $r = 0.53$	The positive slope is stronger in oil than in utilities, but again has the predicted sign.

Read. The second empirical check asks whether equity yields increase with debt-to-equity ratios. They do.

Economic message. This is the empirical counterpart to the paper’s central theoretical claim that leverage raises the cost of equity instead of lowering the overall cost of capital.

5.7 Exhibit 5: tax-adjusted project hurdle rates

Table 5: How taxes alter financing-specific hurdle rates

Financing source	Required return in the paper	Economic message
New common stock	$p_k^S = p_k^\tau / (1 - T)$	The project must cover corporate taxes when financed externally with equity.
Debt	$p_k^D = \frac{p_k^\tau}{1-T} \left(1 - \frac{rT}{p_k^\tau}\right)$	Interest deductibility lowers the hurdle rate relative to new equity.
Retained earnings	$p_k^R = \frac{1-T_d}{1-T_g} \cdot \frac{p_k^\tau}{1-T}$	Personal taxes on dividends versus capital gains make internal funds potentially cheaper than new equity.

Read. Taxes destroy the clean financing irrelevance of the baseline proposition.

Economic message. When capital structure matters in this paper, it matters because of explicit tax wedges, not because debt is intrinsically a value-creating security.

6 Short summary

- The paper defines the cost of capital in a risky world using market valuation rather than subjective utility or certainty-equivalent shortcuts.
- In perfect capital markets, firm value is independent of capital structure: $V_j = X_j/p_k$.
- The average cost of capital is therefore constant within a risk class, while the cost of equity rises with leverage to offset cheap debt.
- The same logic implies an investment rule: accept projects with $p^* \geq p_k$, regardless of financing source.
- Corporate taxes create a debt advantage through the interest tax shield, but the paper argues the quantitative wedge is much smaller than standard doctrine implied.
- The preliminary evidence from utilities and oil companies is broadly consistent with a flat overall cost of capital and a positive relation between leverage and common-stock yields.

7 Conclusion

7.1 Core conclusion

The paper's central claim is that in perfect capital markets the firm cannot create value simply by reshuffling claims among debt and equity. The market values the operating cash-flow stream, not the labels attached to the securities used to finance it.

7.2 Economic intuition

The intuition is simple but powerful. Whenever corporate leverage appears to create value, investors can replicate or undo that leverage on their own balance sheets. As soon as this replication is feasible, any valuation premium for one capital structure over another becomes an arbitrage opportunity and cannot persist in equilibrium.

7.3 Takeaway

This paper is best read as a benchmark theorem. Its force comes from showing exactly when capital structure does *not* matter. Once that benchmark is clear, every meaningful theory of financing must explain which friction—taxes, bankruptcy costs, agency problems, asymmetric information, market segmentation, or something else—breaks the M–M logic.